COLLEGE: **DON BOSCO COLLEGE (Co-Ed), YELAGIRI HILLS**

COLLEGE CODE: **358**

DEPARTMENT: BCA

PROJECT TITLE:

Farm Management System

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ABSTRACT

The Farm Management System is a cloud-based application designed to digitalize and streamline agricultural operations. The system was developed to maintain structured records of farmers, crops, fields, and farming activities in a centralized platform.

The project focuses on creating a customizable data-driven environment where agricultural information can be stored, organized, and analyzed efficiently. It provides modules for managing farmer details, crop information, land records, and activity tracking. In addition, the system includes reporting and dashboard features that help in visualizing farm data for better decision-making.

The application demonstrates how modern cloud platforms can be used to transform traditional agricultural record-keeping into a structured and automated management system.

INTRODUCTION

Agriculture is a vital sector that requires proper planning, monitoring, and record maintenance to ensure productivity and sustainability. In many cases, farm-related information such as land details, crop cycles, and farming activities are maintained manually. This leads to data inconsistency, difficulty in tracking performance, and lack of proper analysis.

To overcome these challenges, the Farm Management System was developed as a centralized digital solution. The system allows users to create and manage structured records for farmers, crops, fields, and farming operations. It enables data entry, relationship mapping, report generation, and dashboard visualization.

The main objective of this project is to design an organized system that improves transparency, reduces manual errors, and provides meaningful insights through reports and graphical representation of data.

PROBLEM STATEMENT

Traditional farm management practices face several issues:

1. Lack of centralized data storage.
2. Difficulty in maintaining accurate farmer and crop records.
3. No structured method to track farming activities.
4. Inability to analyze seasonal crop performance.
5. No visual representation of agricultural data.
6. Data duplication and inconsistencies.

SYSTEM OVERVIEW

The Farm Management System is developed as a modular application consisting of multiple interconnected components. The system allows users to create records, define relationships between data entities, generate reports, and visualize information using dashboards.

The main components include:

- Farmer Management Module
- Field Management Module
- Crop Management Module
- Activity Management Module
- Reporting Module
- Dashboard Module

Each module plays a specific role in organizing and managing farm-related data.

SYSTEM MODULES IN DETAIL

1. Farmer Management Module

This module stores detailed information about farmers. It includes fields such as:

- Farmer Name
- Contact Information
- Address
- Total Land Area
- Registration Date

This module acts as the primary record holder for all farmer-related data. Each farmer record can be linked to multiple land records.

2. Field Management Module

The Field module stores land-related information such as:

- Field Name
- Location
- Soil Type
- Area Size
- Irrigation Type

Each field record is linked to a specific farmer. This helps in organizing land ownership details and tracking productivity per field.

3. Crop Management Module

The Crop module stores details about crops grown in different fields. It includes:

- Crop Name
- Season
- Crop Type
- Sowing Date
- Expected Yield
- Harvest Date

This module allows tracking of crop cycles and seasonal cultivation patterns.

4. Activity Management Module

The Activity module records farm operations such as:

- Irrigation
- Fertilization
- Pest Control

Each activity record contains:

- Activity Name
- Activity Date
- Activity Type
- Status (Completed / Pending)

DATA STRUCTURE AND RELATIONSHIPS

The system is designed using structured data entities that are interconnected through relationships.

Example relationships:

- One Farmer can have multiple Fields.
- One Field can have multiple Crops.
- One Crop can have multiple Activities.

These relationships ensure proper data organization and avoid duplication. They also allow users to view related information easily through linked records.

DATA FIELDS AND TYPES

Various field types were used in the system:

- Text Fields for names and descriptions
- Number Fields for area and yield
- Date Fields for sowing and harvesting
- Picklist Fields for season and activity type
- Lookup Fields for creating relationships
- Formula Fields for automatic calculations

Using different field types ensures data accuracy and structured input.

USER INTERFACE DESIGN

The application interface was customized to improve usability. Page layouts were organized to:

- Display important fields prominently
- Arrange related information systematically
- Provide easy navigation between modules

Tabs were created for each module to allow quick access. The application interface is simple and user-friendly, ensuring smooth data entry and retrieval.

REPORT GENERATION

Reports were created to analyze stored data. These reports help in understanding trends and performance.

Examples of reports:

- Farmers grouped by location
- Crops categorized by season
- Activities based on completion status
- Fields classified by soil type

Reports can be filtered and sorted to generate meaningful insights.

DASHBOARD VISUALIZATION

A dashboard was created to visually represent important farm data. It includes:

- Total number of farmers
- Total number of crops
- Number of active activities
- Completed activities
- Seasonal crop distribution

Charts such as bar graphs, pie charts, and summary metrics were used to provide clear visualization.

The dashboard allows users to monitor farm data at a glance without checking individual records.

VALIDATION AND DATA ACCURACY

To ensure correct data entry, validation rules were implemented. These rules prevent:

- Empty mandatory fields
- Negative values for land area
- Incorrect date entries

Data validation improves reliability and reduces errors.

TESTING PROCESS

The system was tested by:

- Creating sample farmer records
- Linking fields to farmers
- Adding crop details
- Recording activities
- Generating reports
- Viewing dashboard charts

Testing confirmed that all modules functioned correctly and relationships were properly maintained.

ADVANTAGES OF THE SYSTEM

- Centralized data storage
- Structured information management
- Reduced manual errors
- Real-time data visualization
- Improved monitoring
- Easy report generation
- Better decision-making support

LIMITATIONS

- Requires internet access
- Initial learning required for users
- Limited automation in current version

FUTURE ENHANCEMENTS

The system can be enhanced by adding:

- Automated notifications for activities
- Integration with weather forecasting systems
- Mobile accessibility
- Advanced analytics features

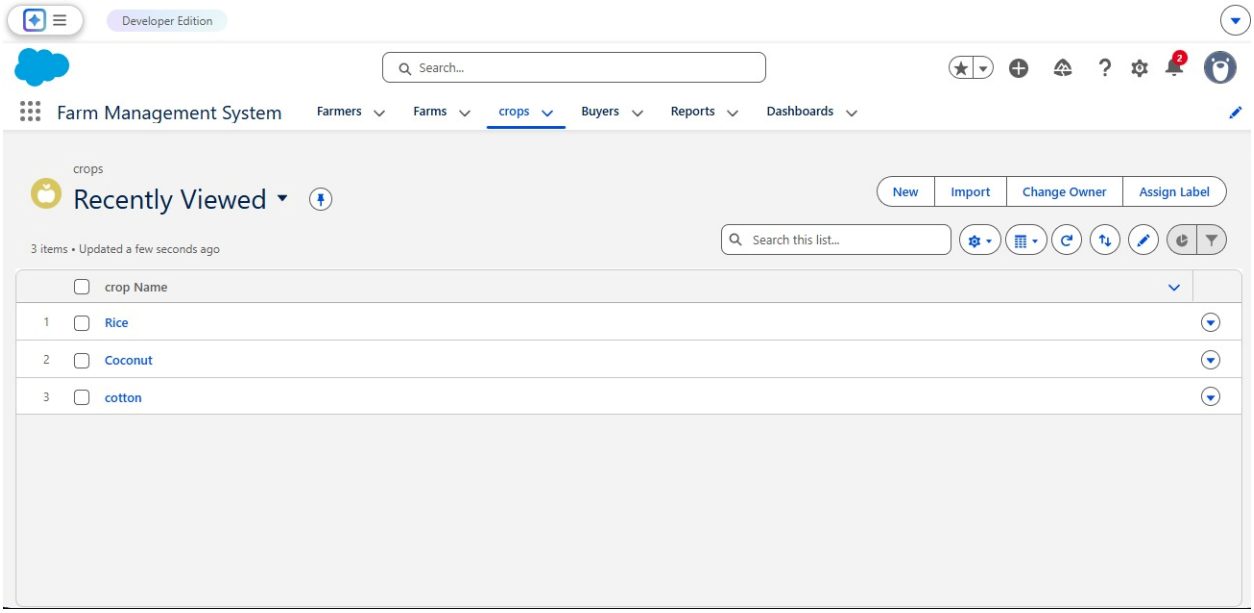
SUMMARY OF THE PROJECT

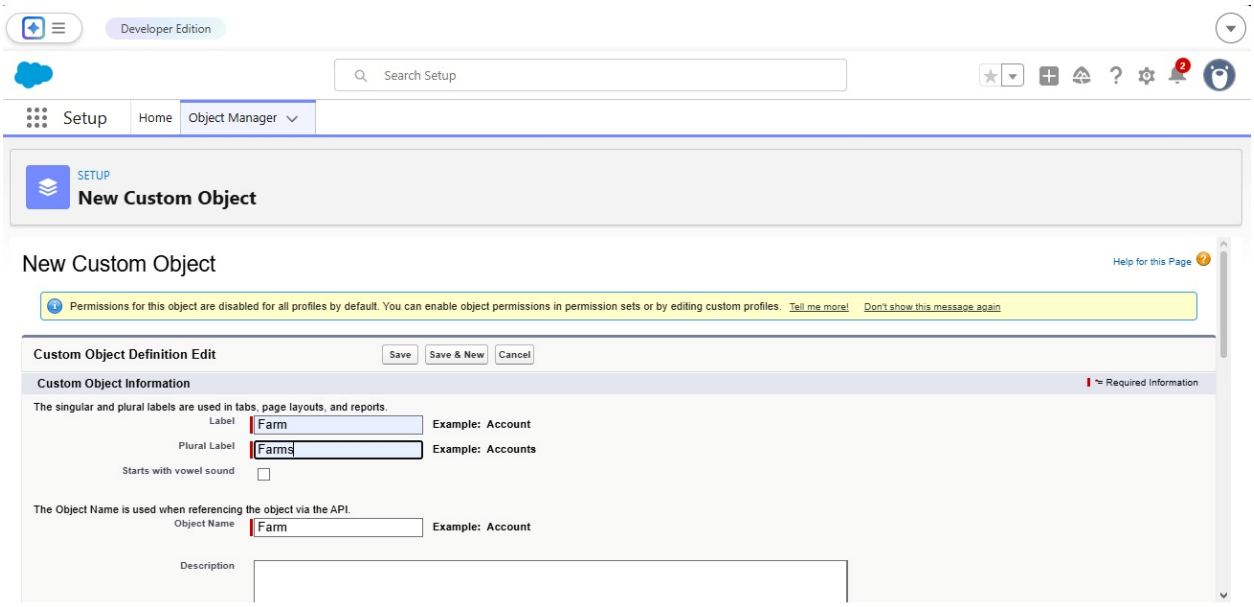
The Farm Management System successfully demonstrates the development of a structured digital application for managing agricultural operations. By organizing farmers, fields, crops, and activities into interconnected modules, the system ensures efficient record maintenance and monitoring.

The reporting and dashboard features provide meaningful insights that support data-driven decision-making. The project highlights how modern cloud-based application platforms can be used to improve agricultural management practices.

This project enhances understanding of application development concepts such as data modeling, relationships, reporting, and dashboard visualization while providing a practical solution for farm management.

SCREENSHOTS:





Developer Edition

Search...

Farm Management System | Farmers | Farms | crops | Buyers | Reports | Dashboards

Farmers

Recently Viewed

0 items • Updated a few seconds ago

New | Import | Change Owner | Assign Label

Search this list...

Developer Edition

Search Setup

Setup | Home | Object Manager

tab

- Feature Settings
  - Analytics
    - Tableau
      - Tableau Embedding
      - Tableau UAF Claims Definition
- User Interface
  - Console Settings
    - Loaded Console Tab Limit
  - Rename Tabs and Labels
  - Tabs

Didn't find what you're looking for? Try using Global Search.

SETUP

Tabs

Custom Tabs

Help for this Page

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs New What Is This?

Action	Label	Tab Style	Description
Edit | Del	Buyers	People	
Edit | Del	crops	Apple	
Edit | Del	Farmers	People	
Edit | Del	Farms	Apple	

What Is This?

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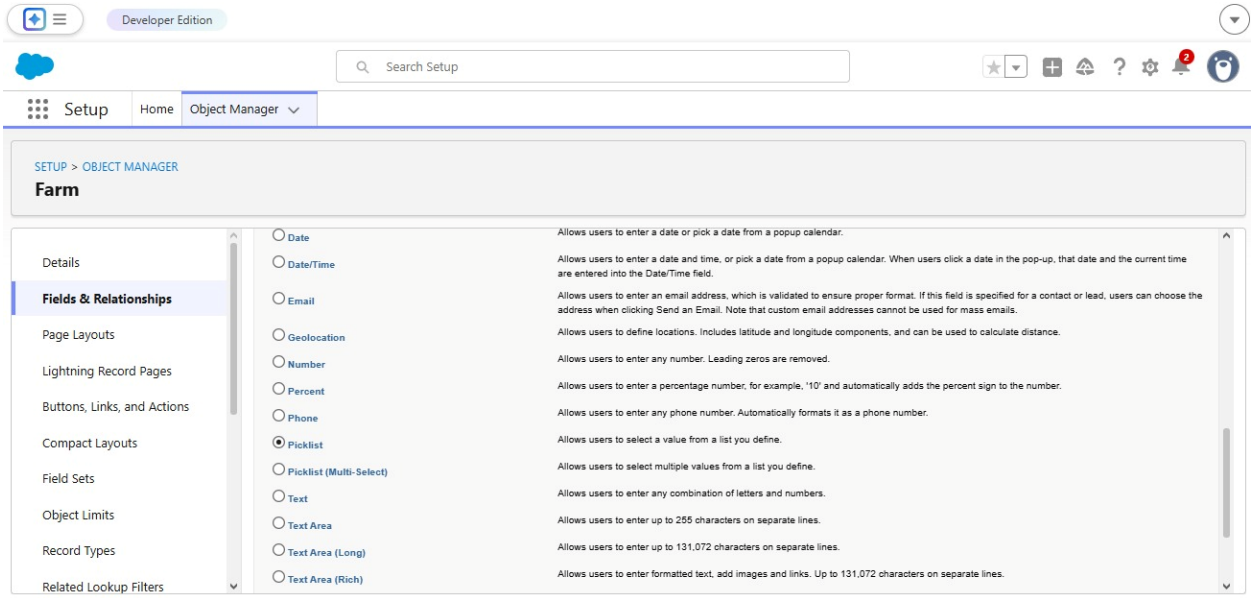
Developer Edition
Search Setup
Setup
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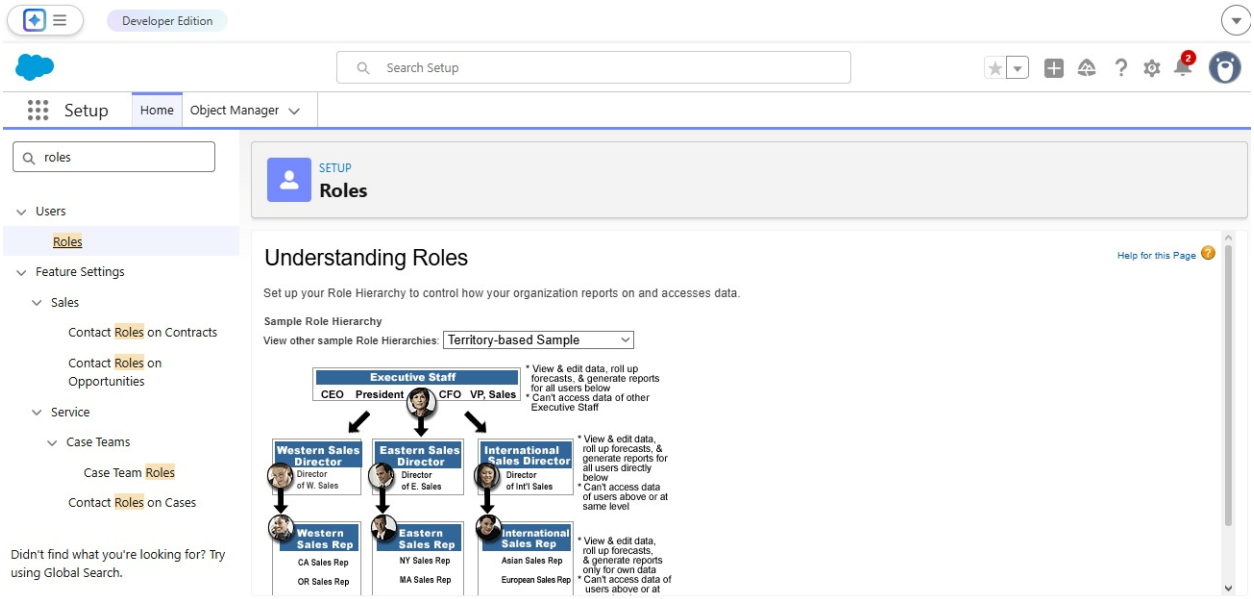
Object Manager
SETUP > OBJECT MANAGER
Farm
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Date
Allows users to enter a date or pick a date from a popup calendar.
Date/Time
Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
Email
Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
Geolocation
Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
Number
Allows users to enter any number. Leading zeros are removed.
Percent
Allows users to enter a percentage number, for example, '10' and automatically adds the percent sign to the number.
Phone
Allows users to enter any phone number. Automatically formats it as a phone number.
Picklist
Allows users to select a value from a list you define.
Picklist (Multi-Select)
Allows users to select multiple values from a list you define.
Text
Allows users to enter any combination of letters and numbers.
Text Area
Allows users to enter up to 255 characters on separate lines.
Text Area (Long)
Allows users to enter up to 131,072 characters on separate lines.
Text Area (Rich)
Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.


Developer Edition
Search Setup
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roles
Users
Roles
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Contact Roles on Opportunities
Service
Case Teams
Case Team Roles
Contact Roles on Cases
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SETUP
Roles
Understanding Roles
Help for this Page
Set up your Role Hierarchy to control how your organization reports on and accesses data.
Sample Role Hierarchy
View other sample Role Hierarchies:
Territory-based Sample
Executive Staff
CEO
President
CFO
VP, Sales
* View & edit data, roll up forecasts, & generate reports for all users below
* Can't access data of other Executive Staff
Western Sales Director
Director of W. Sales
Eastern Sales Director
Director of E. Sales
International Sales Director
Director of Int'l Sales
* View & edit data, roll up forecasts, & generate reports for all users directly below
* Can't access data of users above or at same level
Western Sales Rep
CA Sales Rep
OR Sales Rep
Eastern Sales Rep
NY Sales Rep
MA Sales Rep
International Sales Rep
Asian Sales Rep
European Sales Rep
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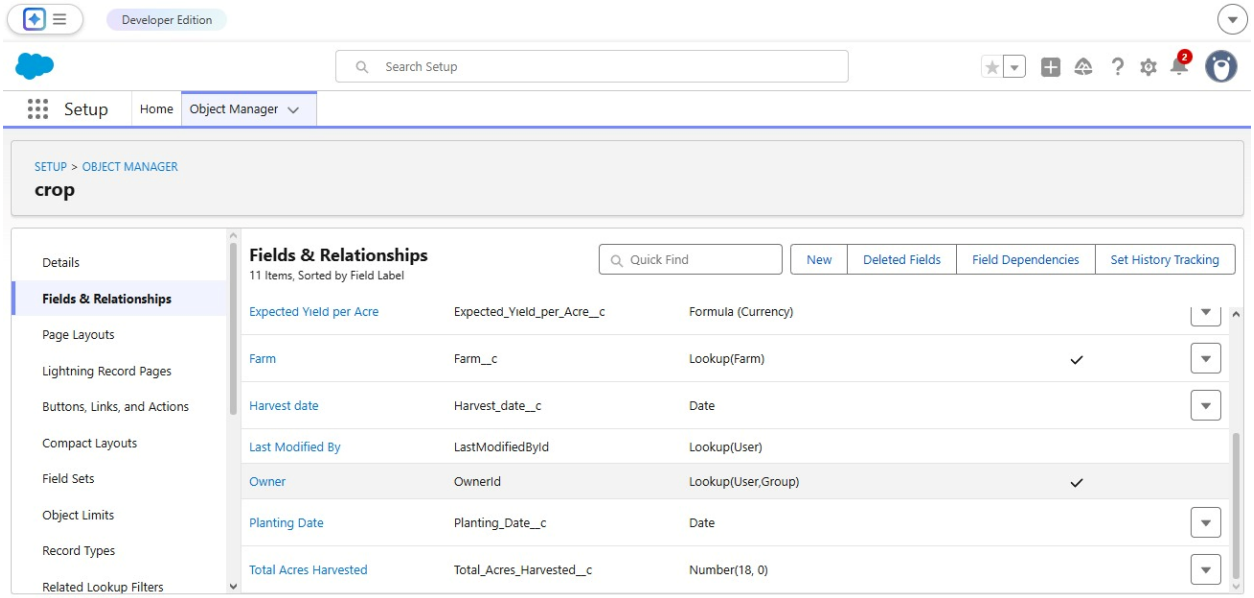
* Can't access data of users above or at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Fields & Relationships
11 Items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
Set History Tracking
Expected Yield per Acre
Expected_Yield_per_Acre__c
Formula (Currency)
Farm
Farm__c
Lookup(Farm)
Harvest date
Harvest_date__c
Date
Last Modified By
LastModifiedById
Lookup(User)
Owner
OwnerId
Lookup(User,Group)
Planting Date
Planting_Date__c
Date
Total Acres Harvested
Total_Acres_Harvested__c
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